STAT 3355.501 Project
Analyzing the deaths of Police officers in America

Group 3
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1 Introduction

This project allows us to put the knowledge we have learned about data manipulation and cleaning to the test as well as providing us the opportunity to use our intuition to analyse the data in a manner that could be expected in the workforce.

The dataset that our group decided to explore concerned information about deaths of police officers in America and special territories under the U.S. government. It contains 22742 observations and 10 variables that detail the manner of death. These variables consists of the officer's title and name(person), the department they were affiliated with and it's location(dept), end of watch information (eow), the cause of death(cause), shortened label for cause of death(cause_short), date in numerical form(date), the year of death(year), the department name without the state (dept_name), the state(state).This dataset was provided by a kaggle page who themselves used data from ODMP(Officer Down Memorial Page). An important note about the data from this organization is that information concerning officer deaths is only recorded if the said officer died in the line of duty or were died off duty but were targeted for their law enforcement affiliation. With that in mind, it is clear that there is a lot of potential within the dataset that could identify trends among officer deaths and raise further questions and create avenues of opportunity for identifying possible causes and solutions.

We decided to utilize this dataset because we felt that police officers are a set of the population that have not received much recognition or concern, particularly by American youth. Officers are a critical part of society and they put their lives on the line to aid and protect us. By working on this data and sharing our findings we hope to raise attention to the state of police officers and the hazards they face within their profession. A major focus of our analysis was revealing relationships between death frequency and a given category such as location, profession, and species (canine or human). Although we initially had a large interest and focus on the relationships between an officer's position and frequency of death, we ended up doing a lot of exploration with the subset of officer's that were K-9s. K-9s are an often overlooked member of police departments and seeing as they are a population that cannot exactly speak for

themselves, it was interesting to explore the relationships surrounding canine deaths. Also, seeing as our dataset goes quite far back in time (1791 at the earliest) there we found many interesting relationships between which states had higher officer casualties throughout the centuries, and based on a general knowledge of U.S. history one could see possible factors that could come into play. We expound on our analysis further in Section 3 of this report.

2 Data Cleaning

The data that we selected was not pretty. Some variables were redundant, and others weren't all that useful the way it was given. So the first step that we took was removing the unnecessary columns. There were two. Dept and cause. Next, we started changing a couple

variable names to something that was easier for us to remember and work with. So for this, we renamed ‘eow’ to ‘day’, and’ date’ to ‘month. You can see that the original names for these vectors would have been confusing to deal with . ‘eow’ means ‘end of watch’ and this column holds lots of information. However, we were only interested in the day of the week the death was caused. Just like ‘eow’, the ‘date’ column had too much needless information that we didnt want to associate with this column, so we changed the name . Next, we changed our ‘day’ column into an ordered factor variable based on the day of the week. This may seem unwarranted, but our ‘day’ column had a lot of different information that could be found in other columns. Only the day of the week of the death was unique to it, so we got rid of everything else. The same problem was found in our ‘month’ column. So we made the variable unique by changing it to an ordered factor variable so that it’s content reflected the name(month), and nothing else. One very important thing we did, and maybe the hardest, was extracting the name of the officer from the ‘position’ column. As the data was given to us, the column ‘position’ held both the name of the officer and the title they had. We thought that by being able to look at the position itself of each officer, we would be able to dive into the data more thoroughly, and find interesting information that we otherwise would not have been able to. Doing all this cleaning to our data made it significantly easier to work on and made our time with it much more efficient.

Another thing we did to present our data in a more organized way was to subset our data for certain plots.l9 By subsetting our data, we could show only specific entries for each column variable. This helped by removing possibly redundant entries on certain variables in order to make a more specific argument based on subsetted plots. For example, for our graphs that showed the relationship between police deaths based on the position or title, we subsetted the first plot to only show only the top 10 most killed positions such as “Sheriff”, and “Police Officer”, even though there were about 60 police positions. We did this because a lot of the different positions were variables such as “Manager” or “Technician” which did not have too many deaths at all and didn't add much to the plot. By Subsetting to show only the top 10 deadliest jobs for police officers, the viewer can much better tell which positions have the highest number of deaths. We then made a second plot concerning police officer deaths based on their positions, but subsetted the state column to show only the top 4 most populous states. This was to get really specific on our second plot, which is now subsetted to show only the top 10 deadliest positions and faceted by the top 4 most populous states.

3 Analysis and Findings

In this section we present the findings of our analysis through the graphics we produced. Our initial tactic when approaching the dataset was referring to some principal questions we drew up when we first looked at the data set such as: were there certain time periods or epochs when overall deaths increased or decreased? How does this information intersect with particular regions of the country (does a particular region experience a spike in deaths during a certain time

period)? Are there any interesting correlations between canine deaths in comparison to human officer deaths? Are there certain officer positions that are more prone to casualties than others? From here, we will expand on the product of our analysis in tackling these questions and exploring possible explanations.

3.1 - Deaths Over the Years

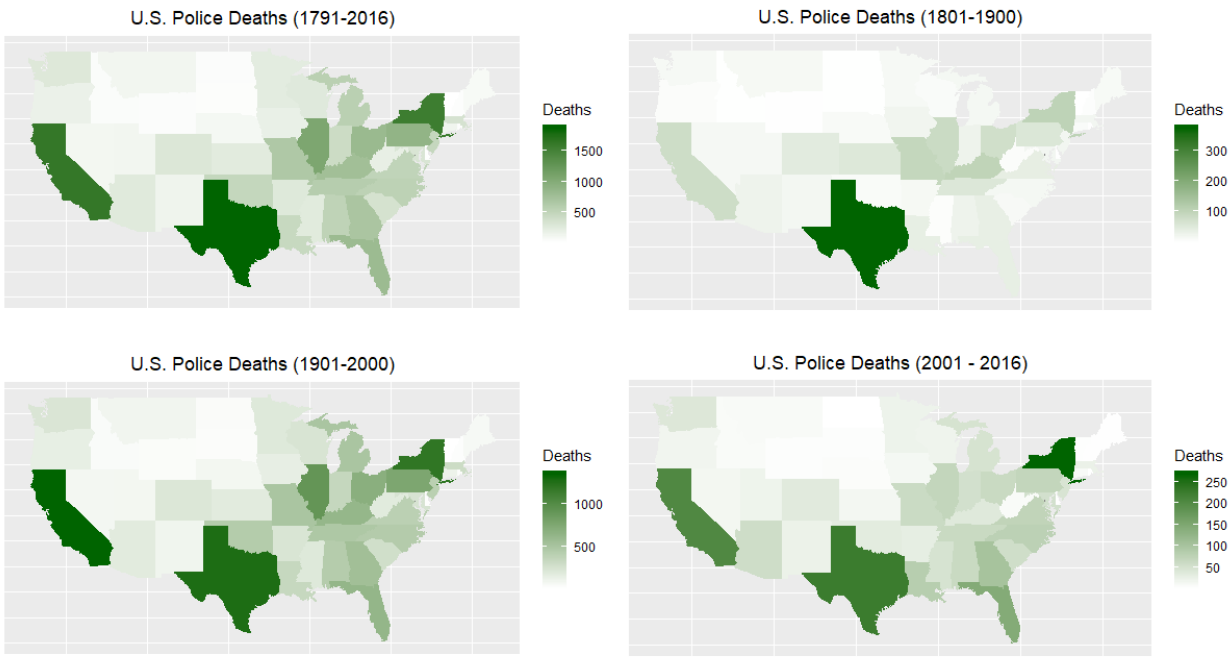

Figure 1: Geographic Frequency of Deaths Over the Centuries

Figure 1 presents an overview of how police officer deaths have changed over time (except Alaska and Hawaii for plotting purposes), with the first graphic in the upper right corner being a summary of deaths over the entire time period recorded within the data set. A factor in why we chose certain time periods concerned the fact the organization from which the data was collected from (ODMP) was created to record police officer deaths initially started from 1971 onwards. They later went back to gather information for years before then, but given the nature of technology and recording in the 18th century as well as the significantly smaller pool of observations from that time period we felt it best to not include it in our analysis and plotting. One of the main questions we can ask upon looking at the graphics are: what is the relationship between death frequency and years and how does it compare to states that are indicated as hotspots over the entire three century period? As can be seen in the 18th century and 19th century map one of the largest changes was that california became one of the focal points concerning officer casualties. Using a basic knowledge of American history, California became a state in the late 19th century. This is probably a contributing factor as far as documentation since many officers probably may not have been counted on American records back then. The

increases in deaths on the west coast from close to none in the 19th century to in the couple hundreds in the 20th century could be attributed to more documentation after western expansion integrating those states into America.

Taking a look at the east coast, one can see that the changes over time are more dynamic than the west coast. We see an overall increase in officer deaths on the east coast which could be attributed to the fact that those areas became quite urbanized and thereby increased in population density more than the rest of the country. An interesting trend that can be noted is that in between the 20th and 21st century in the far midwest region (particularly in the Indiana, Illinois, Ohio region) we see a strong spike in officer deaths in those states in comparison to the 19th and 21st century in those same regions. There are a number of factors that could be involved, a probable one including that other states have become so populous, that it overshadows that region and they in fact still have a consistent frequency of deaths. However, we only have about 20 years of data available to us for the 21st century so further analysis or conclusions are difficult to extract from here.

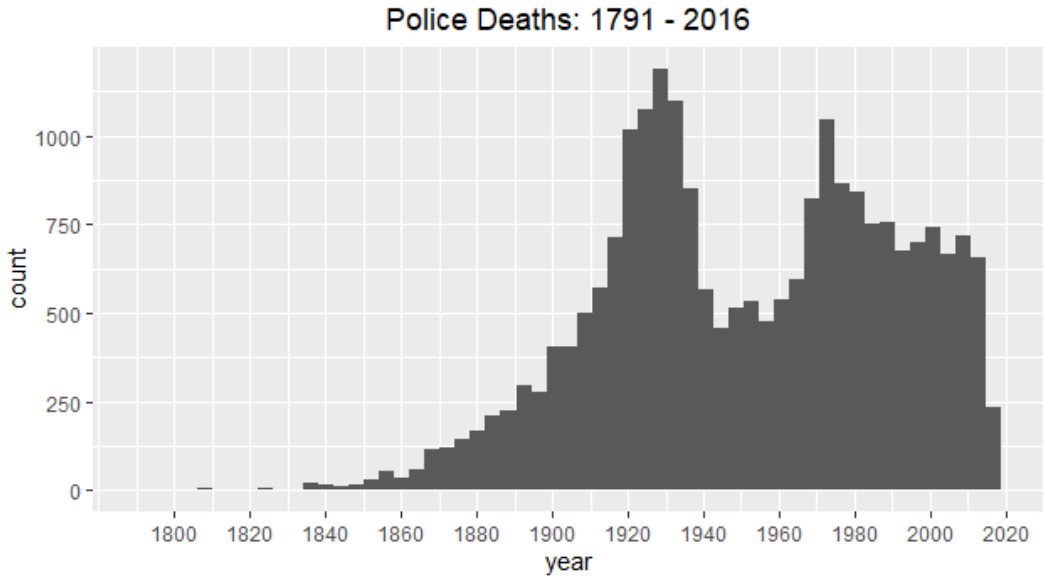

Figure 2: Overall Death Density Over Time

We believed that looking at the geographic intersection of deaths would give us some insight into interesting relationships that could uncover what causes higher frequencies of officer deaths. From Figure 1 the major takeaways we had were that population was an obvious indicator as well as states that have good documentation. In this Figure 2 we take a step back and look at the death count only in reference to the years. The histogram has a bimodal distribution with two peaks, the first being around the year 1930 and the late 1920s and the second being in the 1970s.

Regarding the first peak around the year 1930, one could call attention to the fact that there is a noticeably steady increase leading up to that date. Keeping in mind that America was still a developing country during that early 20th century time frame and the conclusions we gathered from Figure 1 that a prominent factor in police deaths has to due with population

density, it is speculated that due to rising immigrant and urban density at the time that this was the main driver of the trend observed. As to the drop in officer deaths that occur between the 1940s and 1960s, drawing once again on common American history, what is known of that time period is that it was a post world war era where a large portion of the American population was leaving in increasing prosperity. The post world war era angle in particular seems the most influential factor, as that would be an era where crime overall would have decreased due to the trauma of the war and the increased unity among the nation being brought together after an intense period of patriotism. This decreased crime would put officers in less harm through less opportunities of encountering it, however this is as far as we can go with this hypothesis without cross-referencing with other supporting data.

The second peak on the plot around the 1970s was an unexpected find. There are several possible factors that are most likely all contributed to some degree regarding the sustained rise in deaths after that time period as well including the increase in political activism, the rise of gangs in many urban areas, as well as several overseas wars (Vietnam, Gulf, and Iraq wars to name a few). These are all events that cause an increase in intrapopulation violence in a nation and probably created more cases wherein police officers had to get involved. The last interesting find was a sharp drop around the 2010s at the end of our figure, which based on trends highlighted after a previous post war period (1930s) we believe that this drop is also related to similar causes then such as increased patriotism and rising prosperity.

3.2 - On Canine Deaths

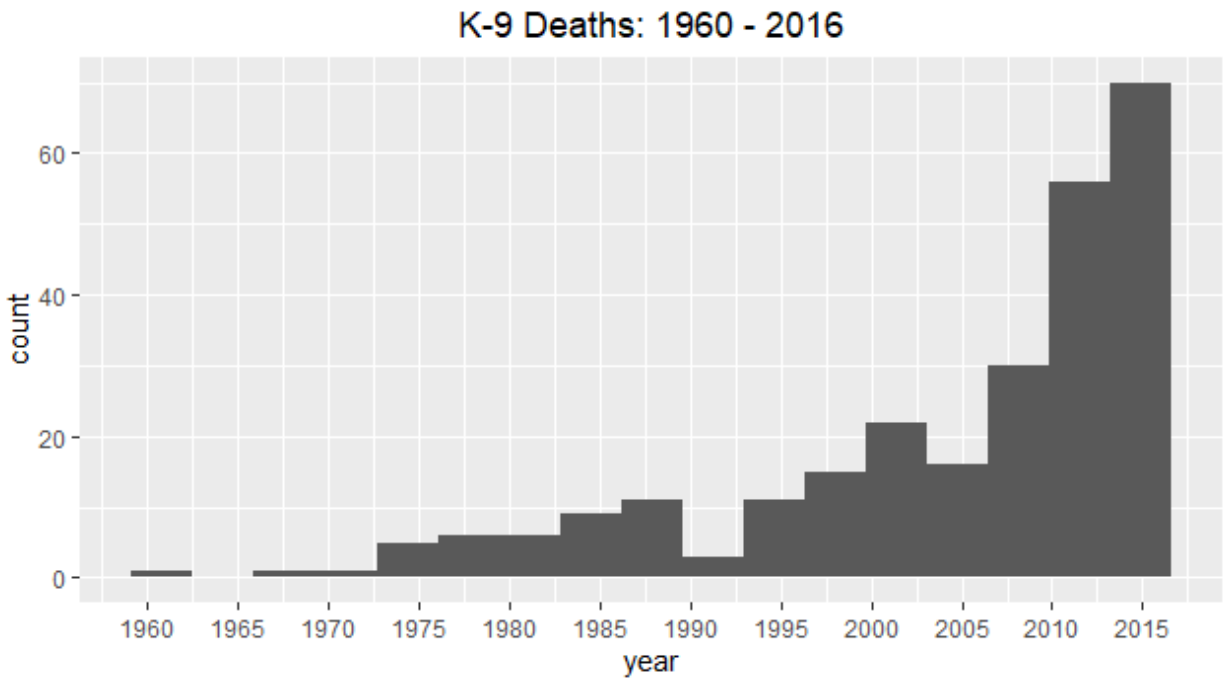

Figure 3: Top causes of canine death

Canines are an essential part of the American police force and help police departments around the nation significantly. And because they are used so much, they contribute to a considerable amount of deaths based on position(More information about position on part 3.3). Figure 4 is a histogram that clearly shows the dramatic rise in canine deaths as the years go from 1960 to 2015. An interesting thing that sticks out is that canine deaths rise at a steady pace, but there is a significant jump from 2005 to around 2013. This could be because of 9/11 and the increasing fight against terrorism. After 9/11, security around the nation increased significantly. I can only assume that the number of dog sniffing dogs were also increased exponentially. So if more dogs were in service during this time period, it would make sense that the death during this period would also be larger. Also, the graph exponentially grew in the 1970s. This may be because the war on drugs in the U.S. began during this time. So the use of canine also increased because of their sense of smell. The dip during the 1990s is something that we do not have the knowledge to explain. When comparing the 'deaths vs time' graph for police officers(humans), the 1990s was not nearly as forgiving. The graph shows that 2015(the last year in the graph) was the year with the most amount of deaths. It wouldn't be far fetched to say that based on this trend, more dogs in service will die in years to come. In the next graphs we talk about some measures to prevent this.

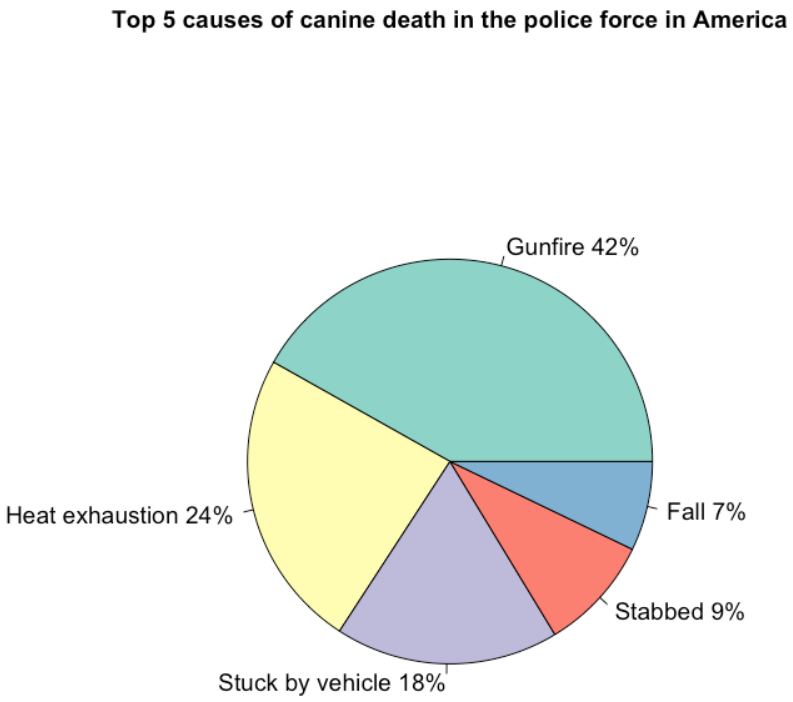

Figure 4: Top causes of canine death

Figure 3 shows the top 5 causes of canine death in America. The pie graph was chosen because it shows the differences in the deaths in a dramatic and easy to see fashion, and gave the graphs in our report more diversity. Unsurprisingly, gunfire caused the most deaths to our police hounds, but what is interesting is that it is the culprit of almost half of the fatalities. This shows that gunfire is not only extremely prevalent in the deaths of human officers, but in canines also, and that bulletproof vest for dogs should not be overlooked. Another interesting thing this graph shows is that there are almost the same number of dogs that died due to a fall as there are that died due to being stabbed. This is concerning because it makes you wonder how these canines are being trained. One misstep can cause these canines to die, and based on the date, it happens enough(7%) to take into consideration. Although, when looking at any data, a healthy dose of skepticism is appreciated, so I will add that the data set that was provided did not specify if the fall was caused by a criminal, or the dog's own clumsiness.

Even with all this taken into account, The most intriguing part of this whole graph is the fact that a quarter of all the canines' death is caused by heat exhaustion. When this was discovered, we decided it needed to be looked into. So we made a new graph

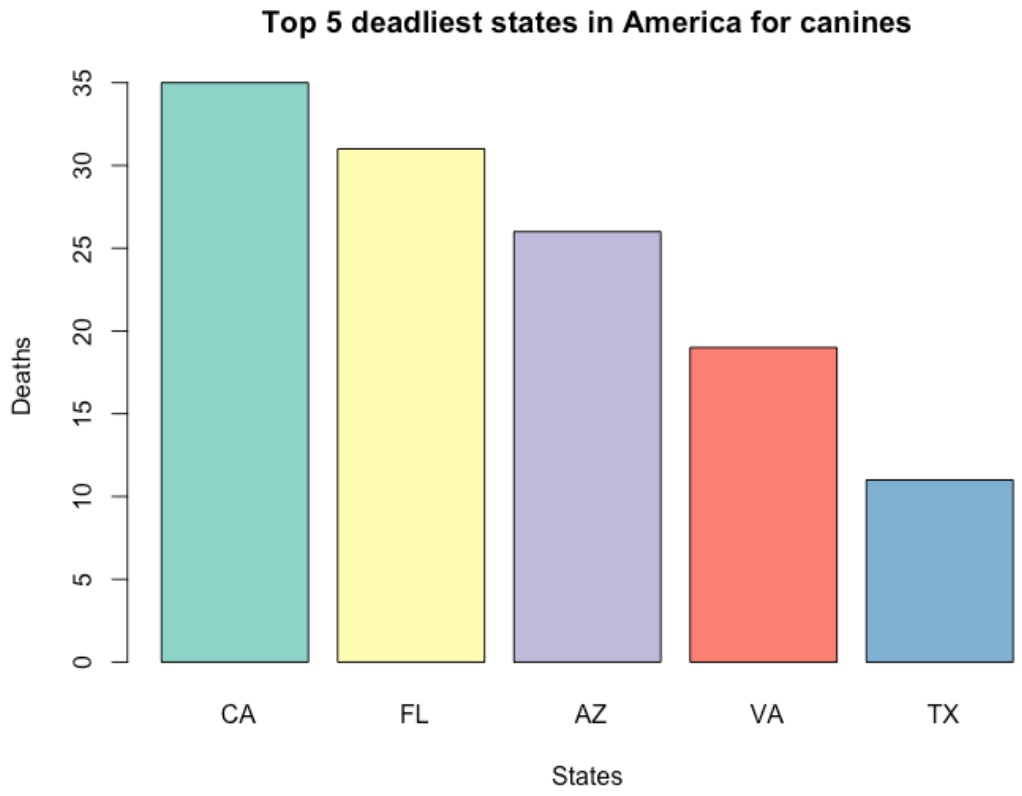

Figure 5: States with most canine deaths

Figure 4 explains so much about our interest in death caused by heat exhaustion. 3 out of the top 5 deadliest states for canines happen to be extremely hot. And even southern California can be just as hot as Florida, Arizona, and Texas. The fact that the second leading cause of death for canines in the police force is heat exhaustion is not something to take lightly. It is almost as much as the combined deaths of stabbing and being stuck by a vehicle. The police departments spend so much time, money, and effort into training these police pals, just for 24% of them to die because they overheated. If we are able to invest in bulletproof vests for these dogs, then the police should take the sun more seriously and provide better cooling and gear for these heros. It also brings up the idea that maybe the type of dogs in the force based on how they fare in the environment where they would need to work should be given a little more thought. One last thing that the figure 4 shows is that 3 out of the 5 states share borders with Mexico. So the high death rates in these states may also be contributed by the fight against drug trafficking.

3.3 - The Significance of an Officer's Title

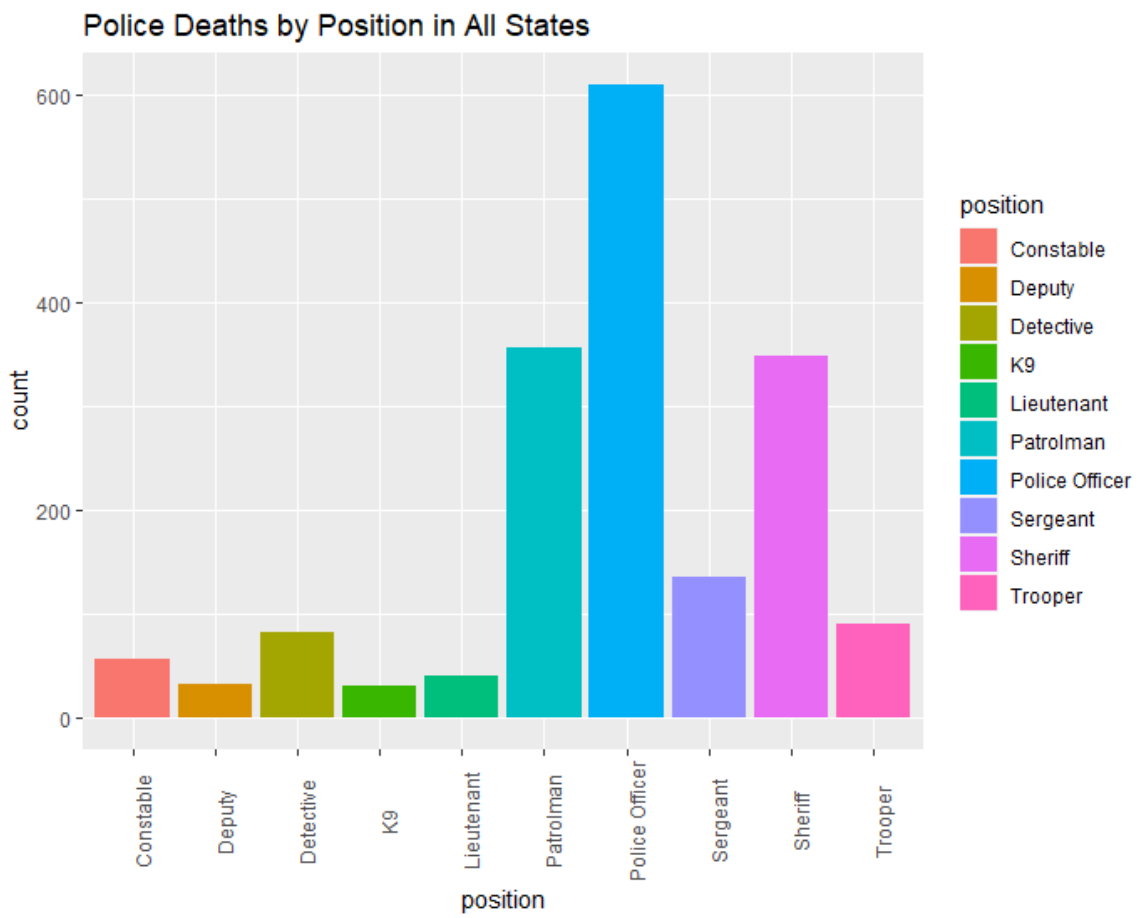

Figure 6: The Effect of Officer Title on Casualty Rates

One thing that was very interesting to us is which positions of the police force recorded the most police deaths over this timeline and why..This is the first plot to display the significance of an officer's position or title in the police force. The plot is Position vs. Police Officer deaths to show the relationship between how deadly each United States police officer position is. The plot is subsetted to show only the top 10 deadliest positions in all the states combined, in order to remove redundant non deadly positions. As you can see from the figure, the deadliest position is “Police Officer”, which is your basic day to day police officer carrying out warrants and arrests, so basically the first policeman on the line, which explains their death toll. That is followed up by “Patrolman” and “Sheriff”. You can see that these three positions are significantly larger than the other 7. A factor that could explain this is that those top 3 positions are positions that typically do more dirty work, or field work that can be very dangerous, such as arresting criminals or dealing with criminals in general.

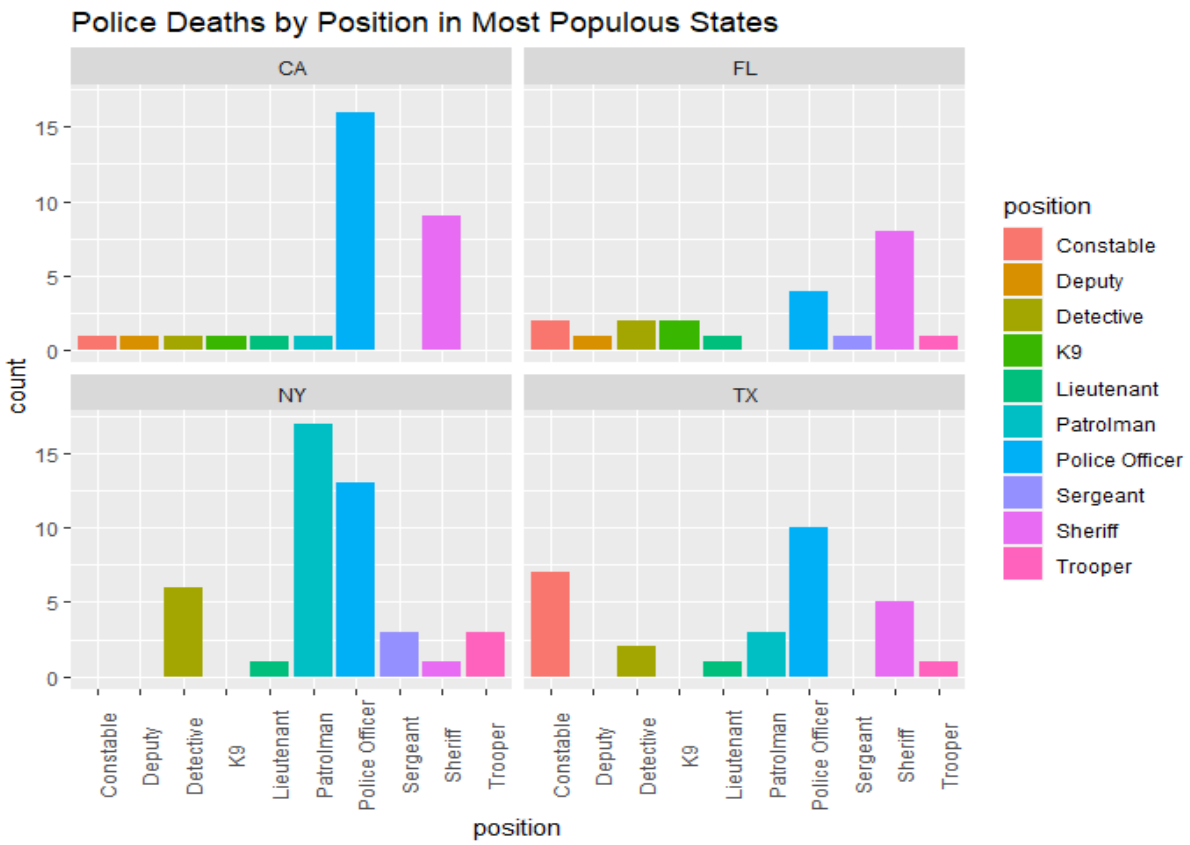

Figure 7: Officer Title Impact by States

The second plot concerning the significance of a police officer's position is very similar to the first one, except it is subsetted to show and facet through only the top 4 most populous states. You can see that there are some key differences between the deadliest position of each state. For California and Tx, the deadliest position is still police officer, which makes sense as they are big states with many large cities and suburbs. For Florida, the deadliest position is "Trooper", which is a state law enforcer who enforces state highway traffic laws. The reason for its deadliness could be due to faulty highways that make it difficult to enforce laws on, and possibly interactions with drug smugglers on Florida highways during the boom in the cocaine industry during the late 20th century. Lastly we have New York, whose deadliest position is "Patrolman". Patrolmen had a very tough job of watching the city at night in New York City, especially in the 20th century when the position was much more needed as technology back then was not as good. The reasons for the differences are probably because each state is in a different region that poses different problems for police that require certain positions of police officers.

3.3 - The Significance of an Officer's Title

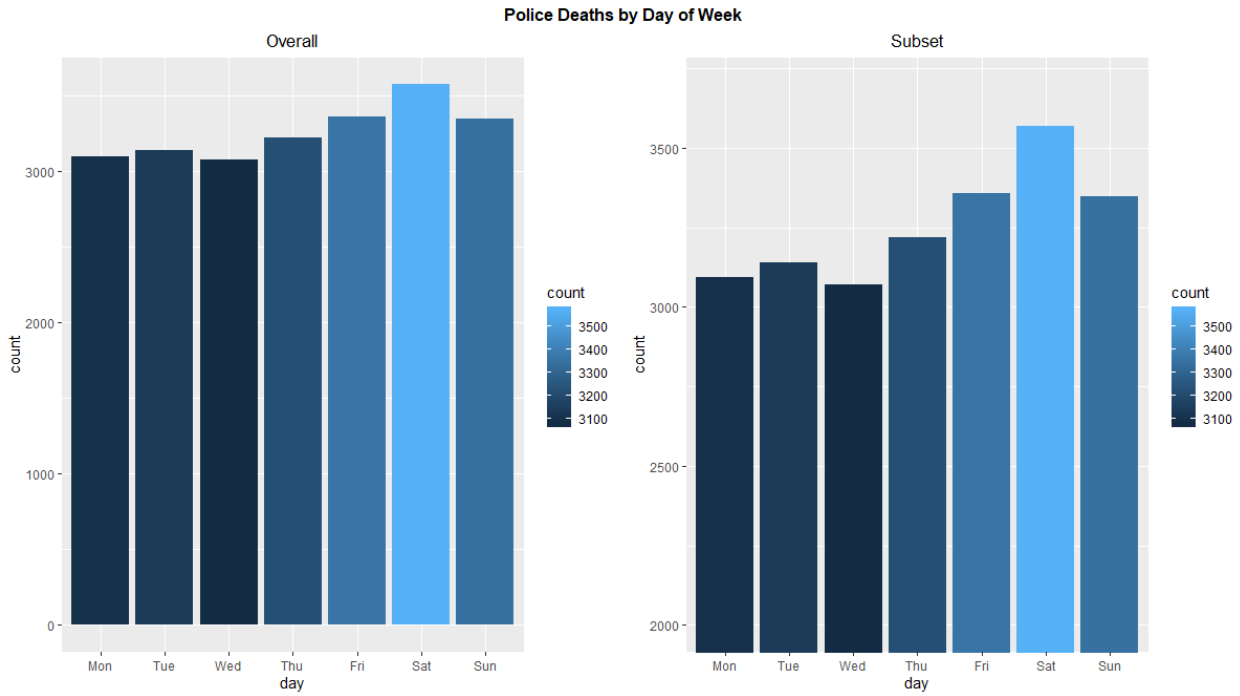

Figure 8: Weekly Police Casualties

Figure 8 here shows the weekly frequencies with which police officers experience fatal casualties. As can be seen in the left figure there all appear to be relatively within the same range except for Friday, Saturday, and Sunday; the weekend. We have the figure on the right that is truncated at 2000 and more casualties so that it can be observed that there is in fact a significant difference between the weekend and the rest of the week. It could be argued by some that it is not the case that the difference is significant, however since our topic concerns the lives of a population that is meant to keep us safe we felt it important to emphasize that it is indeed important. In the figure on the right we can see that Saturday is the deadliest day for police officers as they see an increase of about 5,000 casualties more (around 3500) in comparison with the workdays which are all relatively close to 3000 casualties. Drawing on information about the work and school culture in America and how we generally have those weekend days off as free time and use it as a time to let loose, more people are out on those days as a result. There are more parties among the youth and recreational activities that happen that can put police officers in danger as well as fatal driving accidents that occur more often on the weekend since people happen to be out drinking on those days more often. Overall, since people tend to go out more on the weekend, we believe this is a driving cause of this trend observed. This information could be paired with other data such as map data in figures 1 to triangulate what are the variables that cause an police officer to be most in danger while on duty.

4 Conclusion

In conclusion, the dataset we chose about the death of 22,742 police officers in the line of duty during the time period of 1791-2016, helped our group use our skills of coding and data analysis to come up with interesting results. What led us to choosing this dataset was because we felt that police officers in general are underappreciated, and we were curious about the relationship of certain variables with police officer deaths. But before we could learn anything about our dataset, we first had to clean and reorganize it. This process included moving information in one variable to its own column or removing data from a column variable that was already in another column or was redundant to our arguments, which made our job much easier. After our dataset was cleaned, we began observing the dataset and plotting different relationships about the dataset to come up with some observations and findings.

From playing with the data, we found a lot of interesting information. However, some of it requires basic knowledge about U.S. history to make sense of what the graphs are saying. We found that California changed from having low numbers of officer casualties in the 1800s, to becoming one of the deadliest states in America in the 1900s. We assume it is due to the fact that California became a state in the mid 19th century and began to urbanize at an incredible pace.

We also found that Texas, California, and New York were the three highest police casualty states during this datasets time period, which can be accounted for them being highly populated and urbanized states.

Next we found that the 1920s, 1930s, and 1970s were the deadliest decades for police officers, and that over the whole time period there was a gradual increase in police officer deaths from 1791-1940, with a steep drop off in the 1950s and 1960s, and then a spike in the 1970s, and then decreasing and leveling out in the 1980-2016 time period. This could be accounted for the crime of the roaring twenties and probation, the poverty caused by the great depression, and then the drop off because of World War 2, and then possibly an increase because of the protests of the Vietnam war.

Next we were curious about the K9 units in the police force and found that K9 deaths have gradually increased over time, with the main causes of death being gunfire (42%), heat exhaustion (24%), and struck by vehicle (18%). This could be because K9 units have been used more and more over time to combat drug trafficking and terrorism that has risen in the US. Also the deadliest states for K9s are California, Texas, Arizona, Virginia, and Florida. We assume that since these are mostly southern border states that maybe more K9 are used to combat drug trafficking, and the hot temperatures associated with these states may contribute to the deaths caused by heat exhaustion.

Lastly we found that Police Officer, Patrolman, and Sheriff were the three deadliest positions in the police force. This could be accounted for the fact that these positions are responsible for many dangerous tasks and dealings with criminals, such as arresting criminals and protecting and watching dangerous streets.

Furthermore, more research could be needed to truly delve into what can be done to protect the lives of policemen who protect our streets. Some ideas for future research into this topic include the use of drones and their effectiveness, tracking location of citizens, and better

protective gear. It could also be used to further police officer protection efforts by giving them evidence on which to request hazard pay for certain times that are significantly more dangerous for them than normal. This of course requires some further investigation and cross referencing but provides a possibility for influence for police officers in the future.

Appendix

A - Data Cleaning

```
# importing excel file
```

```
# what is detailed below is that we created a reference data frame named police.deaths
# from which we made the data easier to work with and sometimes used to create other data
# frames for specific plot
police.deaths ← read.csv("clean_data.csv")

# removing redundant columns
police.deaths ← police.deaths[-2]
police.deaths ← police.deaths[-3]

# changing eow to day-of-death & shortening date contain just month information
police.deaths$'eow' ← substr(police.deaths$'eow', 6, 8)
names(police.deaths)[names(police.deaths) == 'eow'] ← 'day'
police.deaths$'date' ← substr(police.deaths$'date', 6, 7)
names(police.deaths)[names(police.deaths) == 'date'] ← 'month'

# changing police.deaths$day to ordered factor variable
police.deaths$'day' ← as.factor(police.deaths$'day')
police.deaths$'day' ← ordered(police.deaths$'day',
                        levels = c("Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"))

# changing police.deaths$months to ordered factor variable
police.deaths$'month' ← as.factor(police.deaths$'month')
levels(police.deaths$month) ← c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep",
"Oct", "Nov", "Dec")

# subset of data that contains U.S. states &  excludes special regions
deaths.states <- filter(police.deaths, state != " PR" &
                    state != " TR" &
                    state != " VI" &
                    state != " MP" &
                    state != " GU" &
                    state != " CZ" &
                    state != " AS")
write.csv(deaths.states, “deaths.states.csv”)
write.csv(police.deaths, “police.deaths.csv”)
# the reason a new file was created was that we did some cleaning in Excel as well to get a
# be able to get a regions variable to making plotting maps easier
```

```
# FIGURE 1
deaths.state ← read.csv("deaths.states.csv")
# load map data and relevant libraries
library(maps)
library(ggplot2)
library(gridExtra)
library(dplyr)
us_map ← map_data("state")

# removing unnecessary information
deaths.state ← deaths.state[-1]
deaths.state ← deaths.state[-1]
deaths.state$state ← as.factor(deaths.state$state)

# separating the 19th, 20th, 21st century
death.19.century ← filter(deaths.state, year >= 1801 & year <= 1900)
death.20.century ← filter(deaths.state, year >= 1901 & year <= 2000)
death.21.century ← filter(deaths.state, year >= 2001 & year <= 2016)

# extracting death frequency
deaths.total ← table(deaths.state$region)
region ← tolower(levels(deaths.state$region))
century.19 ←table(death.19.century$region)
century.20 ← table(death.20.century$region)
century.21 ← table(death.21.century$region)

# creating new df
# used the USArrests data set to quickly create a new df with state names
death.df ← as.data.frame(cbind(region, deaths.total,
                               century.19, century.20, century.21))

# join us_map and death.df
us_map ← left_join(us_map, death.df, by = "region")

# convert factors to numeric
us_map$deaths.total ← as.numeric(as.character(us_map$deaths.total))
us_map$century.19 ← as.numeric(as.character(us_map$century.19))
```

```
us_map$century.20 ← as.numeric(as.character(us_map$century.20))
us_map$century.21 ← as.numeric(as.character(us_map$century.21))

# FIGURE 4
# making subsets of canine deaths only and causes of death
canine_death ← which(police.deaths$canine == T)
cause_canine_death ← police.deaths$cause_short[canine_death]

# passing a data table of causes of death to data table
ccd ← table(cause_canine_death)

# make vector of top 5 deadliest states
top5cause ← ccd[rev(order(ccd))][1:5]
# find out the top 5 causes of death from this barplot
barplot(top5cause, main = "Top 5 causes of canine death", xlab = "States", ylab = "Deaths",
cex.names = .8)

# extracting frequencies of dogs that died by top 5 causes
gun ←length(which(police.deaths$cause_short[canine_death] == "Gunfire"))
heat ← length(which(police.deaths$cause_short[canine_death] == "Heat exhaustion"))
car ← length(which(police.deaths$cause_short[canine_death] == "Struck by vehicle"))
stabbed ← length(which(police.deaths$cause_short[canine_death] == "Stabbed"))
fall ← length(which(police.deaths$cause_short[canine_death] == "Fall"))

# FIGURE 5
# making subsets to extract canine death information by state and passing info to a vector
states_canine_death ← police.deaths$state[canine_death]
scd ← table(states_canine_death)

# FIGURE 6
#Filtering dataset to show only top ten most death stricken occupations for police officers
all.states ← filter(police.deaths, position == c("Trooper", "Lieutenant", "K9", "Patrolman",
"Sheriff", "Police Officer", "Detective", "Deputy", "Sergeant", "Constable"))

# FIGURE 7
#Filtering state variable to show only 4 most populous states, and filtering position variable
states ← filter(police.deaths, state == c(" TX", " FL", " NY", " CA"))
```

```
States_Position ← filter(states, position == c("Trooper", "Lieutenant", "K9", "Patrolman",
"Sheriff", "Police Officer", "Detective", "Deputy", "Sergeant", "Constable"))

# FIGURE 8
deaths$day ← factor(deaths$day, labels = c("Mon", "Tue", "Wed", "Thu", "Fri", "Sat",
"Sun"),
                    levels = c("Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"),
                    ordered = TRUE)
```

B - Code for Section 3 Plots

```
# FIGURE 1
# plot showing the general frequency of police deaths over ~200 years
# relevant libraries
library(ggthemes)
library(gridExtra)

# this plot is going to show the frequency in regards to the entire time span we have data on
all.death ← ggplot(us_map, mapping = aes(x = long, y = lat, group = group)) +
  geom_polygon(mapping = aes(fill = deaths.total)) +
  coord_quickmap() +
  scale_fill_gradient(low = "white", high = "darkgreen", name = "Deaths") +
  labs(title = "U.S. Police Deaths (1791-2016)") +
  theme(plot.title = element_text(hjust = 0.5),
        axis.title.x = element_blank(),
        axis.title.y = element_blank(),
        axis.text.x = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks = element_blank())

# plotting 19th century
death.19 ← ggplot(us_map, mapping = aes(x = long, y = lat, group = group)) +
  geom_polygon(mapping = aes(fill = century.19)) +
  coord_quickmap() +
  scale_fill_gradient(low = "white", high = "darkgreen", name = "Deaths") +
  labs(title = "U.S. Police Deaths (1801-1900)") +
  theme(plot.title = element_text(hjust = 0.5),
        axis.title.x = element_blank(),
        axis.title.y = element_blank(),
        axis.text.x = element_blank(),
```

```
      axis.text.y = element_blank(),
      axis.ticks = element_blank())

# plotting 20th century
death.20 <- ggplot(us_map, mapping = aes(x = long, y = lat, group = group)) +
  geom_polygon(mapping = aes(fill = century.20)) +
  coord_quickmap() +
  scale_fill_gradient(low = "white", high = "darkgreen", name = "Deaths") +
  labs(title = "U.S. Police Deaths (1901-2000)") +
  theme(plot.title = element_text(hjust = 0.5),
      axis.title.x = element_blank(),
      axis.title.y = element_blank(),
      axis.text.x = element_blank(),
      axis.text.y = element_blank(),
      axis.ticks = element_blank())

# plotting 21st century
death.21 <- ggplot(us_map, mapping = aes(x = long, y = lat, group = group)) +
  geom_polygon(mapping = aes(fill = century.21)) +
  coord_quickmap() +
  scale_fill_gradient(low = "white", high = "darkgreen", name = "Deaths") +
  labs(title = "U.S. Police Deaths (2001 - 2016)") +
  theme(plot.title = element_text(hjust = 0.5),
      axis.title.x = element_blank(),
      axis.title.y = element_blank(),
      axis.text.x = element_blank(),
      axis.text.y = element_blank(),
      axis.ticks = element_blank())

# arranging all the plots into one graph
grid.arrange(all.death, death.19, death.20, death.21, ncol = 2)

# FIGURE 2
deaths <- read.csv("police.deaths.csv")
x <- deaths$year
n <- length(x)

# Rice rule
k <- ceiling(2 * n^(1/3))
```

```
# BINS = 57

# overall trend of all officer deaths over time
ggplot(data = deaths, mapping = aes(x = year, y = ..count..)) +
  geom_histogram(bins = k) + # k = 57: Rice Rule
  labs(title = "Police Deaths: 1791 - 2016") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_x_continuous(breaks = c(seq(1800, 2020, by = 20)))

# FIGURE 3
# separating canine deaths
canines ← filter(deaths, canine == TRUE)

# creating bins
x ← canines$year
n ← length(x)
# square-root choice
k ← ceiling(sqrt(n))
# BINS = 17

# histogram for canine death frequency
ggplot(data = canines, mapping = aes(x = year, y = ..count..)) +
  geom_histogram(bins = k) + # Square Root Choice: Bins = 17
  labs(title = "K-9 Deaths: 1960 - 2016") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_x_continuous(breaks = c(seq(1960, 2020, by = 5)))

# FIGURE 4
# color palette
library(RColorBrewer)

# making the pie chart
slices ← c(gun, heat, car, stabbed, fall)
lbls ← c("Gunfire", "Heat exhaustion", "Stuck by vehicle", "Stabbed", "Fall")
pct ← round(slices/sum(slices)*100)
lbls ← paste(lbls, pct) # add percents to labels
lbls ← paste(lbls,"%",sep="") # ad % to labels

# pie chart
```

```
pie(slices, labels = lbls, main = "Top 5 causes of canine death in the police force in America",
    col = brewer.pal(5,'Set3'), cex = 1.2)

# FIGURE 5
# rearranging information so that it will show in the plot ordered
top5states ← scd[rev(order(scd))][1:5]

# barplot showing top 5 deadliest states
barplot(top5states, main = "Top 5 deadliest states in America for canines", xlab = "States", ylab
= "Deaths", col = brewer.pal(5,'Set3'))

# FIGURE 6
#Plotting Position vs. Police Deaths
ggplot(all.states) + geom_bar(mapping = aes(
  x = position, fill = position), position = "dodge") +
  ggtitle("Police Deaths by Position in All States") +
  theme(axis.text.x = element_text(angle = 90))

# FIGURE 7
ggplot(States_Position) + geom_bar(mapping = aes(
  x = position, fill = position), position = "dodge") +
  ggtitle("Police Deaths by Position in Most Populous States") +
  facet_wrap(~(state)) +
  theme(axis.text.x=element_blank()) +
  theme(axis.text.x = element_text(angle = 90))

# FIGURE 8
# creating the graph separating deaths by weekday
overall ← ggplot(data = deaths) +
  geom_bar(mapping = aes(x = day, fill = ..count..)) +
  labs(title = "Overall") +
  theme(plot.title = element_text(hjust = 0.5, size = 13))

# truncating previous graph
subset ← ggplot(data = deaths) +
  geom_bar(mapping = aes(x = day, fill = ..count..)) +
  labs(title = "Subset") +
  theme(plot.title = element_text(hjust = 0.5, size = 13)) +
  coord_cartesian(ylim=c(2000,3700))
```

```
# combining graphs
grid.arrange(overall, subset, ncol = 2,
        top = textGrob("Police Deaths by Day of Week",
                hjust = 0.5,
                gp = gpar(fontface = "bold")))
```

References

[1] “Police Officer Deaths in the U.S”
https://www.kaggle.com/fivethirtyeight/police-officer-deaths-in-the-us